ARJUNA JEE 2027

Lecture -03 | Mathematics

PERMUTATIONS AND COMBINATIONS

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Topics *to be covered*

A **Formula for P & C**

B **Question Practice**

C

D



H.W. Discussion

Find number of solution of following equations:

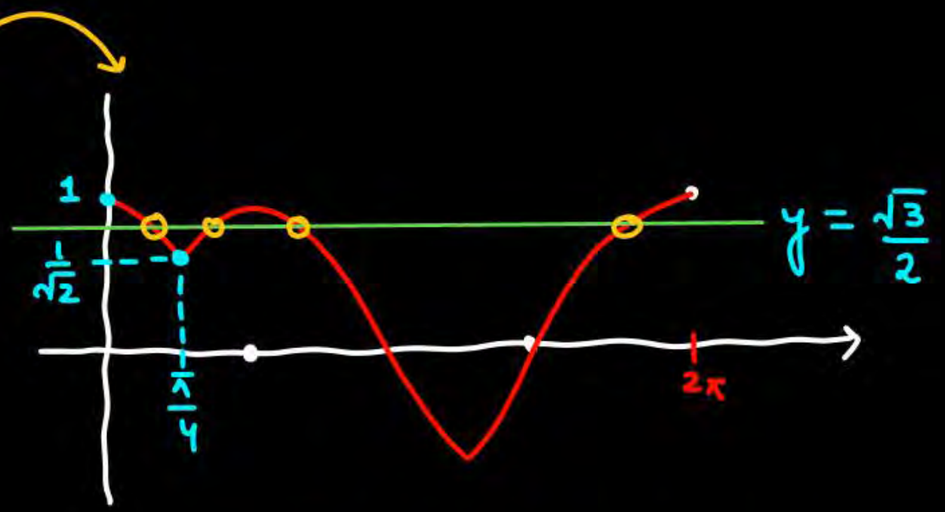
(i) $|\sin \pi x| = \ln x$

(ii) $\max.\{\sin x, \cos x\} = \frac{\sqrt{3}}{2}, x \in [0, 100\pi]$

(iii) $\min.\{\tan x, \cot x\} = \frac{1}{8}, x \in (-2\pi, 2\pi)$

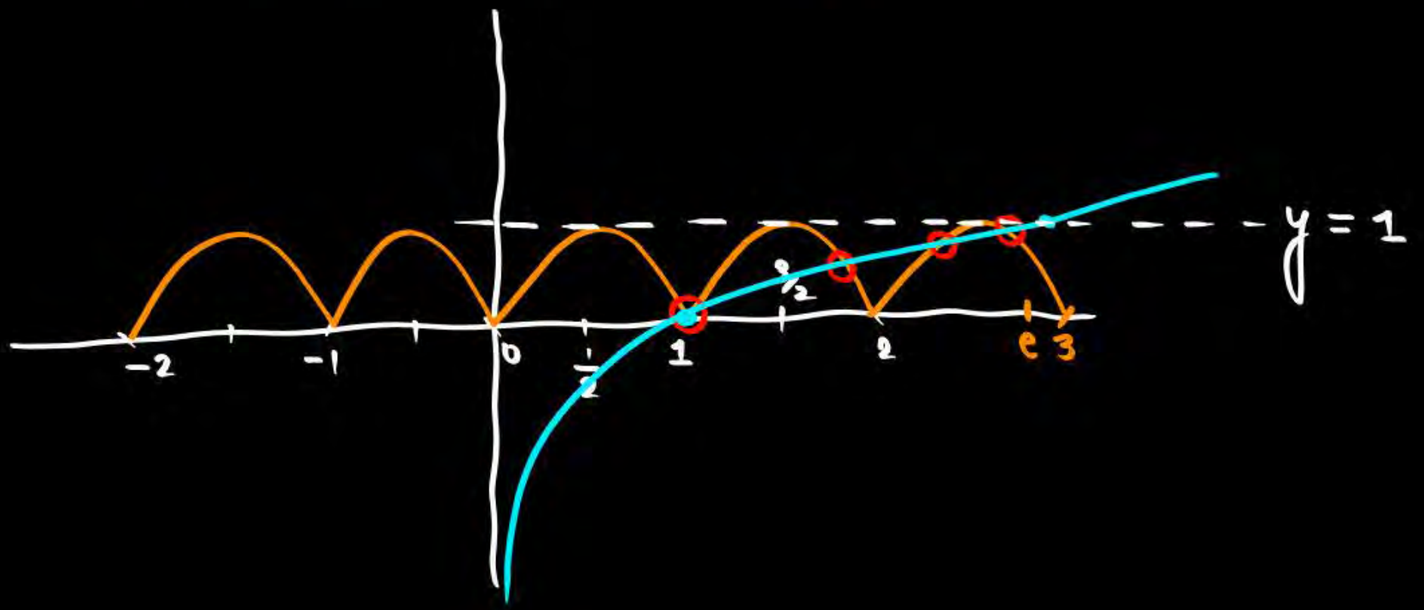
(iv) $2\sin|x| - |x| + 1 = 0$

(v) $3|\cos x| = 5|\sin x|, x \in [0, 3\pi]$

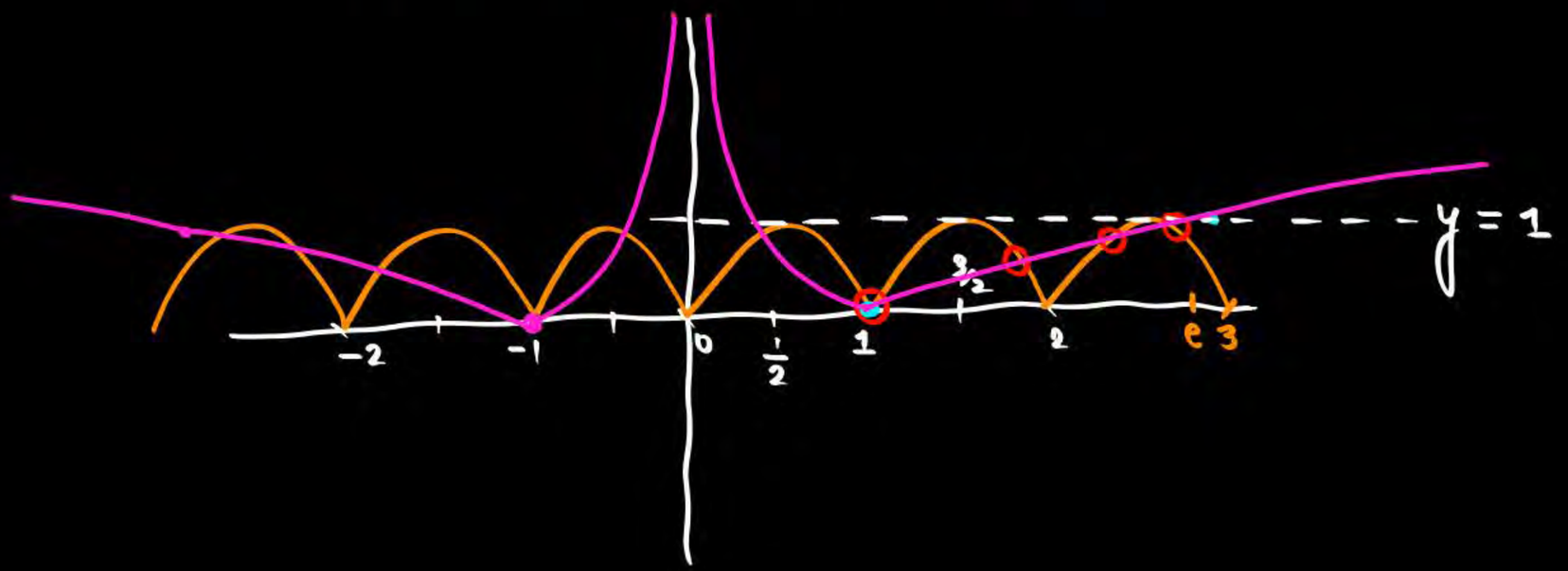


$\text{Ans} = 4 \times 50 = 200.$

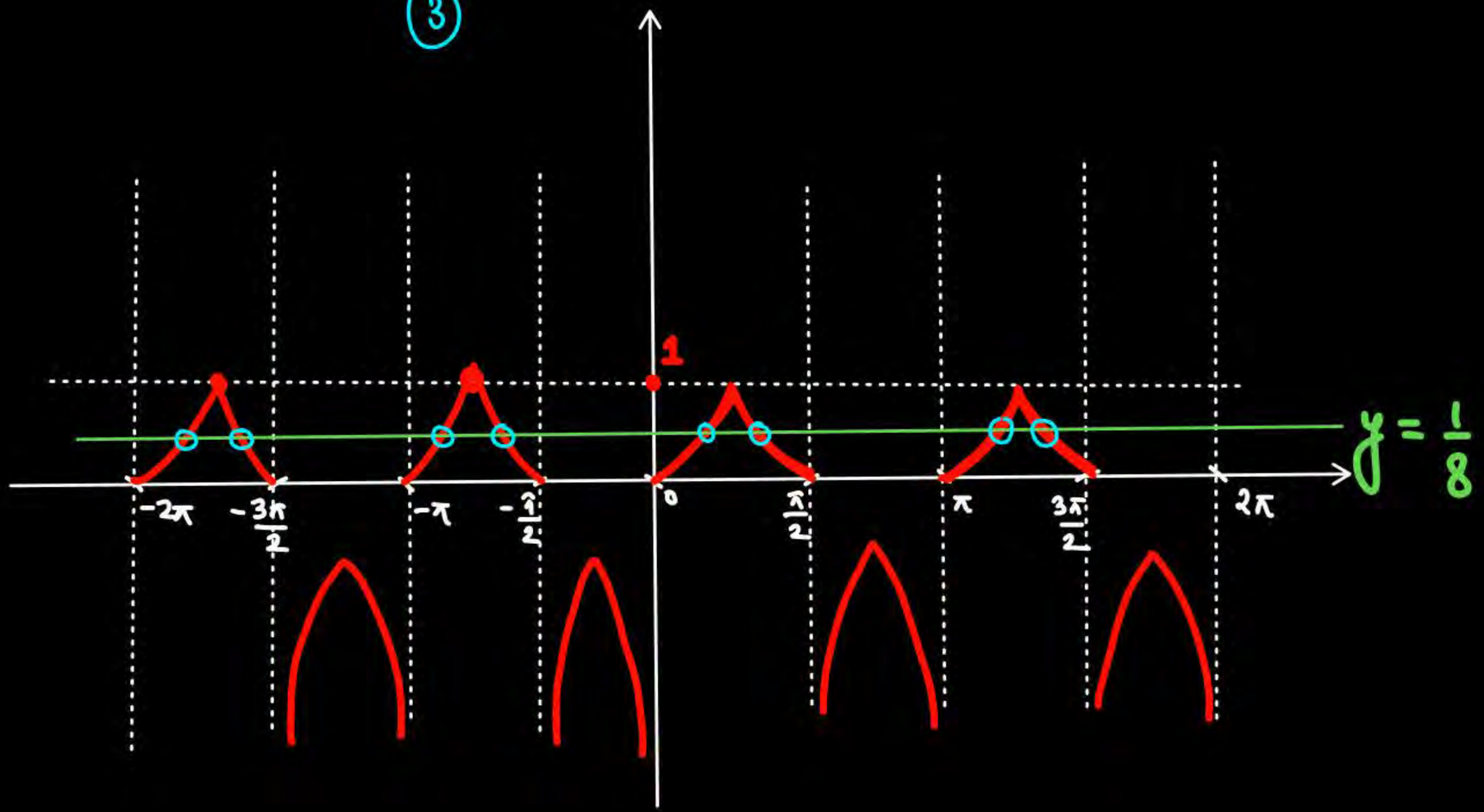
(i) $|\sin \pi x| = \ln x$ → (4)



$|2\sin \pi x| = |\ln|x||$ → (10) SOL^{n}



3



Find number of solution of following equations:

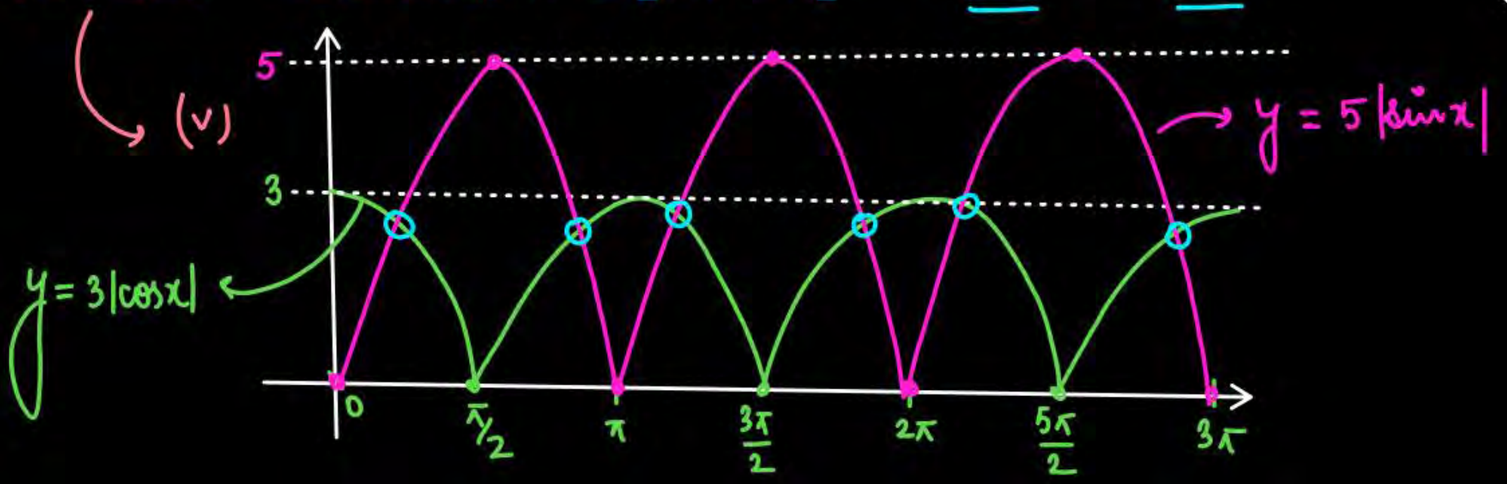
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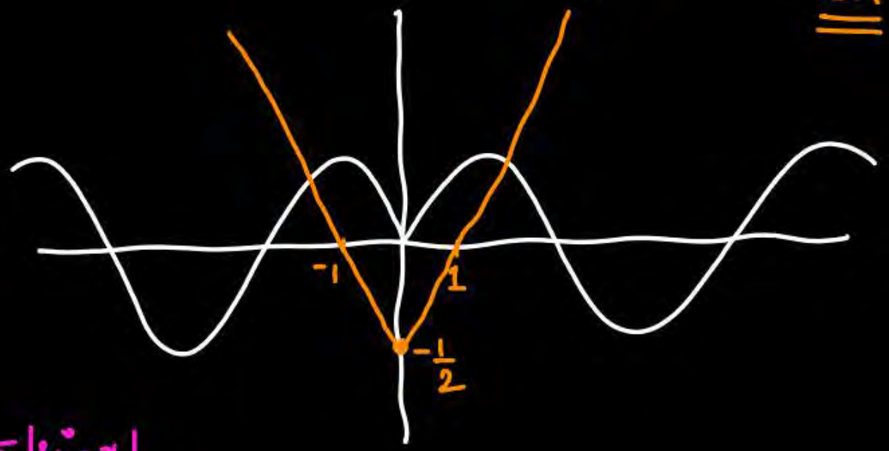
(iv) $2\sin|x| - |x| + 1 = 0$

(v) $3|\cos x| = 5|\sin x|, x \in [0, 3\pi]$ ⇝ 6 soln. 8 soln.



(iv)

$\sin|x| = \frac{|x| - 1}{2}$ → (2) soln.



PERMUTATIONS & COMBINATIONS



DIVISIBILITY RULE

RAR

Divisibility Rule for : 2,3,4,5,6,7,8,9,10,11,25

(i) All those number having their last digit an even number are divisible by 2.

(ii) If the sum of digits is divisible by 3, then number is divisible by 3.

(iii) Divisible by '4' $\Rightarrow$ Last two digits of number is divisible by 4.

(iv) If the last digit of the number is 0 or 5, the number is divisible by 5.

(v) Divisible by '6' $\Rightarrow$ Number should be divisible by 2 & 3 both.

(vi) If the sum of digits is divisible by 9, then the number is divisible by 9.

ex :-

$$abcd = a \times 1000 + b \times 100 + c \times 10 + d$$

$$= a(999+1) + b(99+1) + c(9+1) + d$$

$$= \boxed{(999a + 99b + 9c)} + \boxed{a + b + c + d}$$

$(999a + 99b + 9c)$: div. by 3 OR div. by 9.

$a + b + c + d$: Sum of digits.



DIVISIBILITY RULE

Divisibility Rule for : 2,3,4,5,6,7,8,9,10,11,25

(vii) Divisible by '10' ⇒ Last digit of number is zero.

(viii) Divisible by 11 ⇒ the difference of sum of digits at even and odd places is divisible by 11.

(ix) Whose last two digits are divisible by 25.

(x) If last three digits are divisible by 8 ⇒ Number is divisible by 8.

(xi) Divisibility by 7, double units place digit & subtract it from rest of the number

1 2 3 4 5

\# 3 7 5 4 2 ✗

↑ * ↑ * ↑

Sum of odd place wali digits = 3 + 5 + 2 = 10

Sum of even place wali digits = 7 + 4 = 11

10 − 11 = −1

OR

11 − 10 = 1

7310482 19 ✓

1 2 3 4 5 6 7 8 9

Sum of odd place wali digits = 7 + 1 + 4 + 2 + 9 = 23

Sum of even place wali digits = 3 + 0 + 8 + 1 = 12

23 − 12 = 11 ✓

Question

Find the number of 5 digit integers divisible by 3 which can be formed using the digits {0, 1, 2, 3, 4, 5} without repetition.

Sum = 15 ✓

_ _ _ _ _

C-I : 0, 1, 2, 4, 5.

_ _ _ _ _

$4 \times 4 \times 3 \times 2 \times 1 = 96$.

C-II : 1, 2, 3, 4, 5.

5 4 3 2 1

$= 120$.

216 fnf.

Question [JEE Mains-2022]

The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition, which are multiple of 6, is

A 36

B 48

C 60

D 72

$\{1, 2, 3, 5, 6, 7\}$

$\text{Sum} = 24$

C-I : 1, 2, 5, 6, 7.

C-II :- 1, 2, 3, 5, 7.

even

$4 \times 3 \times 2 \times 1 \times 2$ opt. (2 or 6) $= 48$

$4 \times 3 \times 2 \times 1 \times 1$ opt. $= 24$

Question [JEE (Adv.)-2018]

The number of 5 digit numbers which are divisible by 4, with digits from the set {1, 2, 3, 4, 5} and the repetition of digits is allowed, is _____.

12
24
32
44
52

5 5 5
_ _ _ 1 2 → $5^3 = 125$.
_ _ _ 2 4
_ _ _ 3 2
_ _ _ 4 4
_ _ _ 5 2

$125 \times 5 = 625$ Ans.

RAPID FIRE (FORMATION OF WORDS)

Total number of words that can be formed using "all letters" of the word MACHINE, in which repetition is not allowed such that

(i) No restriction $= 7! = 5040$.

(ii) Relative order of vowels & consonants remains same.

(iii) Word starts with 'A'

(iv) Word ends with a vowel



(i) $_\ _\ _\ _\ _\ _\ _$

$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 7!$

M, A, C, H, I, N, E.

M, C, H, N A, I, E

(ii) C V C V C V C

$4 \times 3 \times 3 \times 2 \times 2 \times 1 \times 1 = 4! \times 3! = 144$.

(iii) A $_\ _\ _\ _\ _\ _$

$6 \times 5 \times 4 \times 3 \times 2 \times 1$

$720 = 6!$

vowel ki jagah vowel & consonant ki jagah conso.

(iv) $_\ _\ _\ _\ _\ _$ V

$6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 3$ opt. $= 6! \times 3$

Question

Consider the word DAUGHTER. How many 4 letter word can be formed from the letter of above word so that each word contain letter G. (R.N.A.)

Sunita.



$G___ = 7 \times 6 \times 5 = 210$

$_\ G\ _\ _ = 210$

$_\ _\ G\ _ = 210$

$_\ _\ _\ G = 210$

$= 210 \times 4$

$= 840.$

Sumit:



Total = $8 \times 7 \times 6 \times 5$

'G' होगा — Req.

'G' नहीं होगा — Not Req.

Req. = $8 \times 7 \times 6 \times 5 - 7 \times 6 \times 5 \times 4$ ($7 \times 6 \times 5 \times 4$)

$= 7 \times 6 \times 5\ (8-4)$

Question

The number of 5 lettered palindromes (with repetition) which can be formed using the letters from the English alphabets, is

26 letters.



MADAM

RACECAR

NITIN

A 26^5

B 26^4

C 26^3

D 26^2

Question

Find the number of words that can be formed using all the letters of the word 'MIRACLE' such that the vowels occupy only the odd positions.



A 576

B 144

C 288

D 720

UNDERSTANDING MEANING OF PERMUTATION & COMBINATION

Class

A, B, C, D

Call any 3 students on the ground



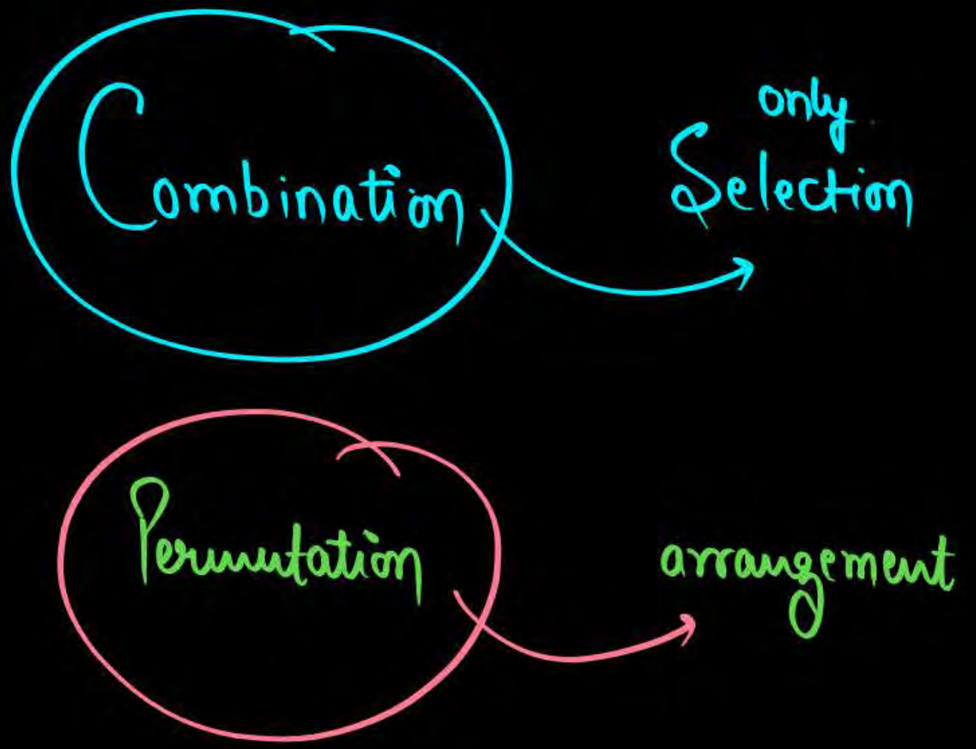
Selecⁿ. OR Comb.

Call any 3 student on the ground & ask them to stand in a line.

Permut. or Arrang.



A B C
A C B
B A C
B C A
C A B
C B A


Combination
only
Selection
Permutation
arrangement

Factorial :-

$n!$ OR $n! = 1.2.3.4.....n$, $n \in N$

$0! = 1$

$1! = 1. = 1$

$2! = 1.2 = 2$

$3! = 1.2.3 = 6$

$4! = 1.2.3.4 = 24$

$5! = 1.2.3.4.5 = 120$

$6! = 1.2.3.4.5.6 = 720$

$7! = 1.2.3.4.5.6.7 = 5040$

#

$15! = 1.2.3.4.....14.15$

$= 14! \times 15$

OR

$n! = n.(n-1)!$

$7! = 7 \times (6!)$

$13! = 13 \times 12!$

$56! = 56.55!$

$n = 1$.

$1! = 1 \times 0!$

$1 = 0!$

FORMULA's FOR P & C

Theorem-01 :

Number of ways of arranging n different things in a line = $n!$

Ex. : Number of ways in which 5 persons can be sited on 5 chairs (in a line) → $5!$

OR

5 different chocolates distributed to 5 boys each get one.

C_1 C_2 C_3 C_4 C_5

A B C D E

FORMULA's FOR P & C

Theorem-02 :

Number of ways of arranging r things taken out of n-different things in a line =

$$^{n}P_{r} = P(n,r) = \frac{n!}{(n-r)!}$$

Ex. : Number of ways in which 5 seats can be filled by 8 persons.

$$^{8}P_{5} = \frac{8!}{3!} \checkmark$$

A, B, C, D, E, F, G, H



$$= \frac{8\times7\times6\times5\times4\times3\times2\times1}{3\times2\times1} = \frac{8!}{3!}$$

$$\# \; {}^{n}P_{n} = \frac{n!}{0!} = n!$$



FORMULA's FOR P & C

Theorem-03 : Process Break

Number of ways of selecting r things out of n-different things = ${}^{n}C_{r}$

$$ {}^{n}C_{r} = \binom{n}{r} = C(n,r) = \frac{n!}{r!\,(n-r)!} $$

A, B, C, D, E, F, G, H

_ _ _ _ _

(Select any 5 out of 8) × (arrange these 5 on 5 seats) = $\frac{8!}{3!}$

(Select any 5 out of 8) × $5!$ = $\frac{8!}{3!}$

(Select any 5 out of 8) = $\frac{8!}{5!\,3!}$

$^{13}C_4 = \frac{13!}{9!\,4!}$

$^{13}P_4 = \frac{13!}{9!}$

C ≡ comb. (selecⁿ.

P ≡ perm. (arrang.

(i) $^nC_0 = \frac{n!}{n!\,0!} = 1.$

(ii) $^nC_1 = \frac{n!}{(n-1)!\,1!} = \frac{n\cdot(n-1)!}{(n-1)!} = n.$

(iv) $^nC_r = {}^nC_{n-r}.$

$\frac{n!}{r!\,(n-r)!} = \frac{n!}{r!\,(n-r)!}$



$^5C_2 = {}^5C_3$

$^{12}C_5 = {}^{12}C_7$

$\frac{1}{(-n)!} = 0$

$\frac{1}{(-n)!}$ is a discontinuous function.

$^4C_7 = 0.$

$$\frac{4!}{7!\,(-3)!} = 0$$

$$\frac{1}{(-3)!} = 0.$$

$$^7C_4 = \frac{7!}{4!\,3!}$$

Question Practice

Question

Using the digits 1, 2, 3, 4, 5, 6, 7, a number of 4 different digits is formed. Find

C_1	C_2
(1) How many number are formed?	(A) 840
(2) How many numbers are exactly divisible by 2?	(B) 200
(3) How many numbers are exactly divisible by 25?	(C) 360
(4) How many of these are exactly divisible by 4?	(D) 40
	(E) 180

Question [JEE Mains-2023]

The number of five-digit numbers, greater than 40000 and divisible by 5, which can be formed using the digits 0, 1, 3, 5, 7 and 9 without repetition, is equal to

A 132

B 120

C 72

D 96

Question [JEE Mains-2022]

Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1, 2, 3, 4, 5 and 6 without repetition of digits. Then the total number of such numbers is _____.

DO IT BY YOURSELF (DIBY)

DIBY-02

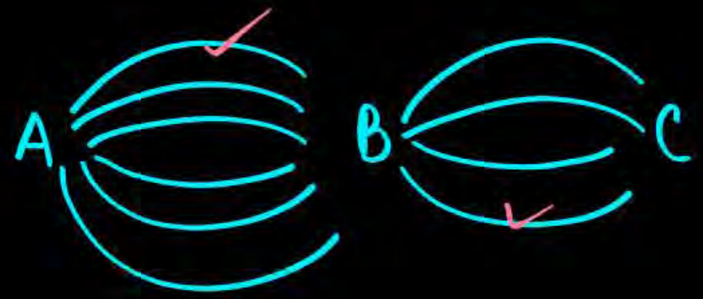
There are 6 roads between A & B and 4 roads between B & C.

(i) In how many ways can one drive from A to C by way of B?

(ii) In how many ways can one drive from A to C and back to A, passing through B on both trips?

(iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.

[Ans. : (i) 24; (ii) 576; (iii) 360]



$6 \times 4 \times 3 \times 5$

DIBY-04

How many natural numbers are their from 1 to 1000 which have none of their digits repeated.

[Ans. : 738]

{1, 2, 3, 4, 5, 6, 7, 8, 9, 0}

ex :-

3 ✓

25 ✓

728 ✓

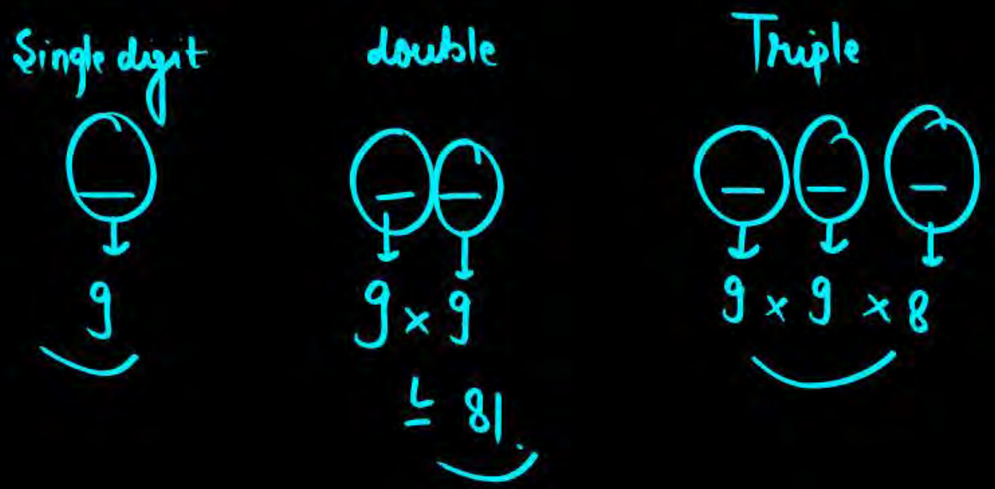
136 ✓

⋮

22 ×

100 ×

737 ×



DIBY-05

How many positive integers less than 10^5 can be written with only the digits 7 and 9 ?

[Ans. : 62]

DIBY-06

(a) How many new words can be formed by using all the letters of the word 'MIRACLE'?

(b) How many words starts with a vowel and ending with a consonant.

(c) How many words starts with a consonant and ends with consonant.

[Ans. : (a) 7! – 1; (b) 1440; (c) 1440]

DIBY-07

Find the number of 10-digit numbers such that the product of any two consecutive digits in the number is a prime number.

[Ans. : 2048]

DIBY-08

Find number of ways in which 7 different colours in a rainbow can be arranged, if green is always in the middle.

[Ans. : 720]

DIBY-09

Find the number of natural numbers between 100 and 1000 such that atleast one of their digits is 7 .

[Ans. : 252]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians